

Manuscript Revision Plan and First Pass Edits

Let's tackle this systematically. First, we'll sharpen the Abstract—the most-read section—so it's concise, quantitative, and mirrors your paper's structure. Then we'll outline edits for each major section, refine figure/table callouts and captions, and ensure a seamless flow. Once the text is solid, we'll draft pseudocode and layer in teaching notes on R, Python, and machine learning.

1. Revised Abstract

Purpose

To define genomic drivers of brain metastasis (BM) in breast cancer and assess how receptor subtype modulates CNS dissemination and survival.

Experimental Design

We studied 1,019 patients in the AACR GENIE BPC Breast Cancer v1.2 cohort (2013–2018) with next-generation sequencing and ≥ 2 years clinical follow-up. Clinical and genomic features were compared between patients with and without BM. We modeled BM risk by cumulative incidence (death as competing event) and estimated overall survival (OS) via Kaplan–Meier analysis.

Results

336/1,019 (33.0 %) developed BM. Those with BM were younger (median 49 vs 54 years; $p < 0.001$) and enriched for HER2-positive (35 % vs 22 %) and triple-negative subtype (28 % vs 14 %; both $p < 0.001$). Twelve-month cumulative incidence of BM was 15 % in triple-negative versus 4 % in other subtypes. TP53, ERBB2, CDK12, and MYC alterations were independently associated with BM (adjusted subhazard ratios 1.8–3.5; all $q < 0.05$). Median OS from sequencing was 61 months in HER2+, 54 months in HR+/HER2–, and 18 months in TNBC (log-rank $p < 0.001$).

Conclusions

TP53, ERBB2, CDK12, and MYC define a genomic signature of CNS tropism in breast cancer. Incorporating molecular profiling into risk stratification may guide early CNS surveillance and inform targeted prevention strategies.

Rationale

- Grouped into Purpose, Experimental Design, Results, Conclusions.

- Added key statistics (medians, percentages, p-values) to quantify findings.
- Used consistent terminology for receptor subtypes.
- Emphasized competing-risk modeling for BM.

2. Section-by-Section Roadmap

1. Introduction• Open with incidence/mortality of BM in breast cancer.

- State the gap: limited genomic characterization of CNS tropism.
- Conclude with study objectives and hypotheses.

2. Materials and Methods• Cohort assembly (Figure 1): bullet-point inclusion/exclusion, with a PRISMA-style flowchart illustrating patient counts at each step.

- Clinical variables (Table 1): list covariates, measurement timing, and summary statistics methods (e.g., chi-square, Wilcoxon).
- Genomic data processing: clear pipeline from VAF/MAF import to binary alteration calls; note institution-specific panels.
- Statistical analyses:• Descriptive (Table 1).
- Cumulative incidence (Fine–Gray).
- Kaplan–Meier OS (log-rank).
- Multivariable competing-risks regression for BM (subdistribution hazards).
- Cox PH for OS.
- Multiple-testing correction.

3. Results• Cohort description (refer to Figure 1, Table 1).

- Risk of BM: present cumulative incidence curves by subtype (Figure 2) with 12- and 36-month estimates.
- Genomic associations: forest plot of subhazard ratios for significant genes (Figure 3A), mutation prevalence bar plot (Figure 3B).
- Overall survival: Kaplan–Meier curves by subtype (Figure 4) and hazard ratios in Table 2.

4. Discussion• Summarize key findings.

- Mechanistic interpretation (TP53, ERBB2, CDK12, MYC).
- Clinical implications (CNS surveillance, targeted prevention).
- Strengths (large cohort, multi-institutional).

- Limitations (retrospective, variable imaging).
- Future directions.

5. Figures & Tables • Figure 1: Flowchart of inclusion/exclusion.

- Table 1: Baseline characteristics (mean±SD or median[IQR]; n [%]).
- Figure 2: Cumulative incidence curves (competing risk).
- Figure 3A: Forest plot of genome-BM associations.
- Figure 3B: Bar chart of mutation prevalence.
- Figure 4: Kaplan–Meier OS by subtype.
- Table 2: Multivariable model results (subhazard & hazard ratios).

3. Next Steps

- Confirm which section you'd like to refine next: Introduction, Methods, Results, or Discussion.
- Provide any specific sentences or paragraphs you find clunky or want to trim.
- After text edits, we'll embed pseudocode into a Supplementary Methods section that mirrors your pipeline and helps reproducibility.

Once the narrative is polished, I'll deliver:

- High-level, language-agnostic pseudocode with clear inputs/outputs.
- Annotated teaching notes on R→Python translation and core machine-learning concepts tailored to your background.

Let me know where you'd like to dive in!